Windows Registry Forensics Analysis

Memory evidence examined with Volatility 2.6

Exercises 1–5 | MemoryDump_Lab1.raw | Win7SP1x64

05

Registry artifacts correlated into one evidence chain

SAM • SOFTWARE • NTUSER.DAT

The investigation moves from identification to correlation

01	**Identify**	Confirm the memory image and select the operating-system profile.
02	**Extract**	Recover account, startup and application-execution artifacts.
03	**Interpret**	Separate expected activity from entries that require scrutiny.
04	**Correlate**	Link the user account, executed script and persistence evidence.

Exercise 1 establishes the correct analysis profile

```
C:\Volatility2>volatility_2.6_win64_standalone.exe -f "C:\Users\rileyworth3\Downloads\MemoryDump_Lab1.raw" imageinfo
Volatility Foundation Volatility Framework 2.6
INFO    : volatility.debug    : Determining profile based on KDBG search...
          Suggested Profile(s) : Win7SP1x64, Win7SP0x64, Win2008R2SP0x64, Win2008R2SP1x64_23418, Win2008R2SP1x64, Win7SP
1x64_23418
                     AS Layer1 : WindowsAMD64PagedMemory (Kernel AS)
                     AS Layer2 : FileAddressSpace (C:\Users\rileyworth3\Downloads\MemoryDump_Lab1.raw)
                      PAE type : No PAE
                           DTB : 0x187000L
                          KDBG : 0xf800028100a0L
          Number of Processors : 1
     Image Type (Service Pack) : 1
                KPCR for CPU 0 : 0xfffff80002811d00L
             KUSER_SHARED_DATA : 0xfffff78000000000L
           Image date and time : 2019-12-11 14:38:00 UTC+0000
     Image local date and time : 2019-12-11 20:08:00 +0530
```

Finding

The first suggested profile was Win7SP1x64. This profile was used for the subsequent Volatility 2 plugins.

Evidence shown

The command, Suggested Profile(s), 64-bit address-space layer and image timestamp are retained in the screenshot.

Exercise 2 identifies the local accounts stored in the SAM

```
C:\Volatility2>volatility_2.6_win64_standalone.exe -f "C:\Users\rileyworth3\Downloads\MemoryDump_Lab1.raw" --profile=Win
7SP1x64 hashdump
Volatility Foundation Volatility Framework 2.6
Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
SmartNet:1001:aad3b435b51404eeaad3b435b51404ee:4943abb39473a6f32c11301f4987e7e0:::
HomeGroupUser$:1002:aad3b435b51404eeaad3b435b51404ee:f0fc3d257814e08fea06e63c5762ebd5:::
Alissa Simpson:1003:aad3b435b51404eeaad3b435b51404ee:f4ff64c8baac57d22f22edc681055ba6:::
```

Recovered accounts

Administrator	RID 500
Guest	RID 501
SmartNet	RID 1001
HomeGroupUser$	RID 1002
Alissa Simpson	RID 1003

Meaning

SmartNet provides the identity anchor used to interpret the UserAssist execution path.

Training evidence extracted from `MemoryDump_Lab1.raw`. The displayed hashes belong to isolated laboratory accounts and are shown solely to demonstrate SAM credential extraction.

Exercise 3 finds no suspicious entry in the tested Run key

```
C:\Volatility2>volatility_2.6_win64_standalone.exe -f "C:\Users\rileyworth3\Downloads\MemoryDump_Lab1.raw" --profile=Win
7SP1x64 printkey -o 0xfffff8a001032010 -K "Microsoft\Windows\CurrentVersion\Run"
Volatility Foundation Volatility Framework 2.6
Legend: (S) = Stable   (V) = Volatile

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Registry: \SystemRoot\System32\Config\SOFTWARE
Key name: Run (S)
Last updated: 2019-12-04 14:11:26 UTC+0000

Subkeys:

Values:
REG_SZ        VBoxTray        : (S) C:\Windows\system32\VBoxTray.exe
```

Observed entry

VBoxTray — C:\Windows\system32\VBoxTray.exe

Assessment

Consistent with VirtualBox Guest Additions. No related autorun entry for st4G3$$1.bat was observed in this tested key.

Exercise 4 reveals one program-launch record that requires follow-up

```
C:\Volatility2>volatility_2.6_win64_standalone.exe -f "C:\Users\rileyworth3\Downloads\MemoryDump_Lab1.raw" --profile=Win
7SP1x64 userassist > userassist.txt
Volatility Foundation Volatility Framework 2.6

C:\Volatility2>findstr /i ".exe .bat .ps1 .cmd" userassist.txt
REG_BINARY    %windir%\system32\calc.exe :
REG_BINARY    %windir%\system32\SnippingTool.exe :
REG_BINARY    %windir%\system32\mspaint.exe :
REG_BINARY    %windir%\system32\magnify.exe :
REG_BINARY    {6D809377-6AF0-444B-8957-A3773F02200E}\Microsoft Games\Solitaire\solitaire.exe :
REG_BINARY    D:\VBoxWindowsAdditions.exe :
REG_BINARY    %windir%\explorer.exe :
REG_BINARY    D:\VBOXWINDOWSADDITIONS-AMD64.EXE :
REG_BINARY    {6D809377-6AF0-444B-8957-A3773F02200E}\Oracle\VirtualBox Guest Additions\VBoxDrvInst.exe :
REG_BINARY    %windir%\system32\taskhost.exe :
REG_BINARY    %windir%\system32\msdt.exe :
REG_BINARY    C:\Users\SmartNet\Downloads\Firefox Installer.exe :
REG_BINARY    C:\Users\SmartNet\AppData\Local\Temp\7zS4F6F72A8\setup-stub.exe :
REG_BINARY    %windir%\system32\cmd.exe :
REG_BINARY    %windir%\system32\OptionalFeatures.exe :
REG_BINARY    C:\Users\SmartNet\Downloads\DumpIt_2451523934.exe :
REG_BINARY    C:\Users\SmartNet\Downloads\DumpIt\DumpIt.exe :
REG_BINARY    %windir%\system32\NOTEPAD.EXE :
REG_BINARY    C:\Users\SmartNet\Desktop\st4G3$$1.bat :
REG_BINARY    C:\Users\SmartNet\Downloads\winrar-x64-580.exe :
REG_BINARY    {6D809377-6AF0-444B-8957-A3773F02200E}\WinRAR\Uninstall.exe :
REG_BINARY    %windir%\system32\calc.exe :
REG_BINARY    %windir%\system32\SnippingTool.exe :
REG_BINARY    %windir%\system32\mspaint.exe :
```

Expected activity

calc.exe, mspaint.exe, explorer.exe, Firefox Installer.exe and DumpIt.exe provide normal or lab-related context.

Notable activity

C:\Users\SmartNet\Desktop\st4G3$$1.bat

Why it stands out

- Batch script format
- Unusual filename
- User Desktop location
- Contents not yet recovered

Exercise 5 connects identity, execution and persistence evidence

SAM

SmartNet exists

Local account recovered from the SAM hive.

USERASSIST

Script launch recorded

SmartNet profile records st4G3$$1.bat.

RUN KEY

No match found

Only VBoxTray was observed in the tested key.

Correlation conclusion

The UserAssist evidence is consistent with st4G3$$1.bat being launched from the SmartNet profile, but it does not establish successful execution, malware activity or persistence.

Combined evidence supports a cautious conclusion

Source	Observed evidence	Interpretation	Confidence
Image profile	Win7SP1x64 selected	Correct profile for subsequent plugins	High
SAM	SmartNet account recovered	Identity anchor for user activity	High
SOFTWARE Run	VBoxTray.exe only	No suspicious entry in tested key	Moderate
UserAssist	st4G3$$1.bat recorded	Potentially suspicious launch record	High
Overall	**Artifacts correlate to SmartNet**	**Compromise remains unconfirmed**	**Moderate**

EVIDENCE CHECKLIST

Four screenshots provide a clear, auditable evidence trail

01	**Profile identification**	imageinfo with Win7SP1x64 visible
02	**Account recovery**	hashdump with usernames and RIDs
03	**Autorun inspection**	printkey Run output with SOFTWARE address
04	**Execution history**	filtered userassist showing st4G3$$1.bat

Each embedded screenshot retains the command and the key result needed to support the forensic conclusion.

Next steps should test the suspicious script directly

1 **Recover and inspect**

Extract st4G3$$1.bat and review its commands safely.

2 **Hash and compare**

Calculate SHA-256 and check approved reputation sources.

3 **Correlate processes**

Review pslist, pstree, psscan and cmdscan for related execution.

4 **Check communications**

Use netscan and timeline evidence to identify external activity.

5 **Expand persistence review**

Inspect additional autorun locations, services and scheduled tasks.

Priority

Recover the batch file before making a final malware determination.

The evidence is useful, but its limits must remain explicit

What the analysis establishes

- A compatible Windows profile
- Local account names and RIDs
- One tested autorun value
- UserAssist evidence linked to SmartNet

What it does not establish

- The contents or intent of the batch script
- Whether malware executed successfully
- Complete persistence coverage
- Attribution to a human operator

Analytical lesson

Registry artifacts become stronger when correlated, but defensible reporting separates observation, inference and conclusion.

One suspicious UserAssist record, not a confirmed compromise.

The Win7SP1x64 profile was selected from Volatility’s suggested profiles but was not independently validated using an additional profile-validation method.

Recommended decision: classify as potentially suspicious and continue investigation.